

# Chi (Bruce) Cheng

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## PROFESSIONAL SUMMARY

Software Engineer with hands-on experience building AI systems, agentic pipelines, and full-stack applications. Proven track record shipping production backend services, optimizing ML inference pipelines, and deploying cloud-native systems at scale. Pursuing an MS in Software Engineering at CMU; seeking Summer 2026 SWE internships.

## EDUCATION

### Carnegie Mellon University, Silicon Valley

Master of Science, Software Engineering. GPA: 3.6.

Mountain View, CA

Dec 2026

Relevant Coursework: AI in Industry, Gen AI in Practice, Machine Learning in Software Engineering, Foundation of Computer Systems, Functional Programming

### University of California San Diego

Bachelor of Science, Mathematics-Computer Science. GPA: 3.6.

Apr 2025

Relevant Coursework: Design & Analysis of Algorithms, Advanced Data Structures, Computer Vision, Machine Learning, Recommendation Systems

## EXPERIENCE

### AI Product Developer / Product Manager

Sep 2024 – Jun 2025

Helpport AI

- Built backend systems for an AI-driven call assistant across mortgage, healthcare, insurance, and government sectors using Python, Docker, and CI/CD, reducing latency to 20ms.
- Redesigned the intent-matching pipeline, migrating from Dialogflow to a Transformer-based system using Gemini 2.0 and Vertex AI, cutting inference cost by 6x (\$0.002 to \$0.00031 per match).
- Deployed an AI dialogue flow system using FastAPI, improving response accuracy and cutting handling time by 35%.
- Implemented backend APIs for real-time agent guidance, reducing development cycles from 14 days to 5 days.
- Conducted A/B testing and performance analysis, improving AI model accuracy by 35% and increasing sales conversion by 15%.

### Software Development Engineer Intern

Jul 2024 – Sep 2024

Convooloo

- Developed an AI medical chatbot using FastAPI, LangChain, React, and Google Cloud, identifying 50+ common diseases.
- Implemented RAG-based knowledge retrieval using Terraform and MySQL, reducing average response time from 2.5s to 1.5s.
- Led backend integration with LangServe and LangGraph using FastAPI and NestJS, streamlining service deployment.

## PROJECTS

### NightyNight <https://nightynight-1.onrender.com/>

Apr 2025 – Present

Independent AI Project

- Built a full-stack AI story generator with React/TypeScript frontend and FastAPI backend over SSE for streaming.
- Designed a LangGraph multi-agent pipeline parallelizing chapter writing across LLM nodes with age-adaptive prompting.
- Integrated ElevenLabs Flash v2.5 TTS with one-shot generation (up to 40,000 chars) across 7 narrator voices.

### Parking Spot Locator <https://psl.fogx.link>

Aug 2025 – Dec 2025

CMU / BOSCH Research Collaborator

- Built a Bosch-sponsored locator using vision-language models and CLIP embeddings for spatial reasoning from sensor data.
- Optimized semantic search pipeline, reducing parking query latency from 45s to 15s (3x speedup).

### Research Assistant Agent <https://github.com/bruceche-cmu-F25>

Jan 2026 – Mar 2026

CMU Course Project

- Built a LangGraph AI agent with RAG over 30 academic papers, deployed on Google Kubernetes Engine with auto-scaling.

### Capitawise (2nd Place, \$7,000 Prize)

Mar 2024 – Jun 2024

Franklin Templeton Hackathon

- Led full-stack development of an AI banking chatbot using GPT-4o, Node.js, Flask, and React, cutting response time by 40% and improving resolution from 68% to 85%.

## SKILLS

**Languages & Backend:** Python, Java, C/C++, JavaScript, TypeScript, Go, FastAPI, Flask, Node.js, Express.js, MySQL, MongoDB

**Cloud & DevOps:** AWS, GCP, Docker, Kubernetes, Terraform, CI/CD, GKE

**AI / ML & Frameworks:** LangGraph, LangChain, RAG, LLM APIs, OpenAI API, Vertex AI, PyTorch, TensorFlow, Hugging Face, vLLM, Milvus

**Frontend & Tools:** React, Next.js, Streamlit, Figma, Git, Agile/Scrum